



Confidence-Aware Ensemble and Long-Word Refinement for Artistic Text Recognition

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Introduction

Task: word-level recognition in **artistic text images**.

Problem: decorative fonts, curved layouts, object-like characters, clutter, low text/background contrast and distortions.

Motivation: few studies explore how complementary recognizers can be effectively combined for artistic text recognition.

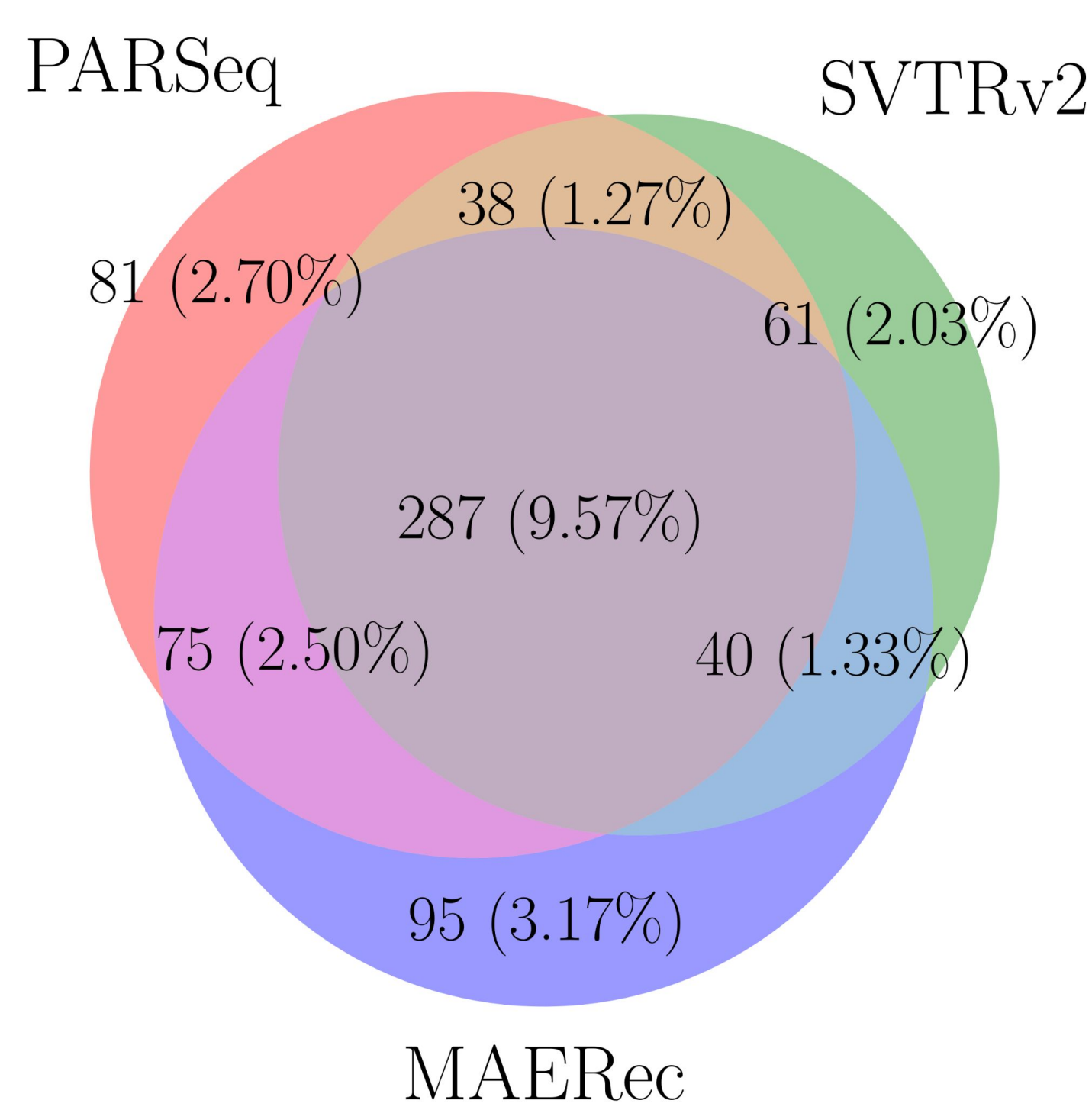


Figure 1. Error overlap among the selected models on Test-A.

Method

Fine-tuning: all recognizers are adapted using the official 6,000-image training split.

Confidence-aware fusion: the ensemble focuses on character positions where the recognizers disagree.

Long-word refinement: words with at least nine characters may undergo sequence alignment and lexicon-guided correction.

- **Needleman-Wunsch alignment:** aligns outputs to handle insertions and deletions before lexicon-guided correction.

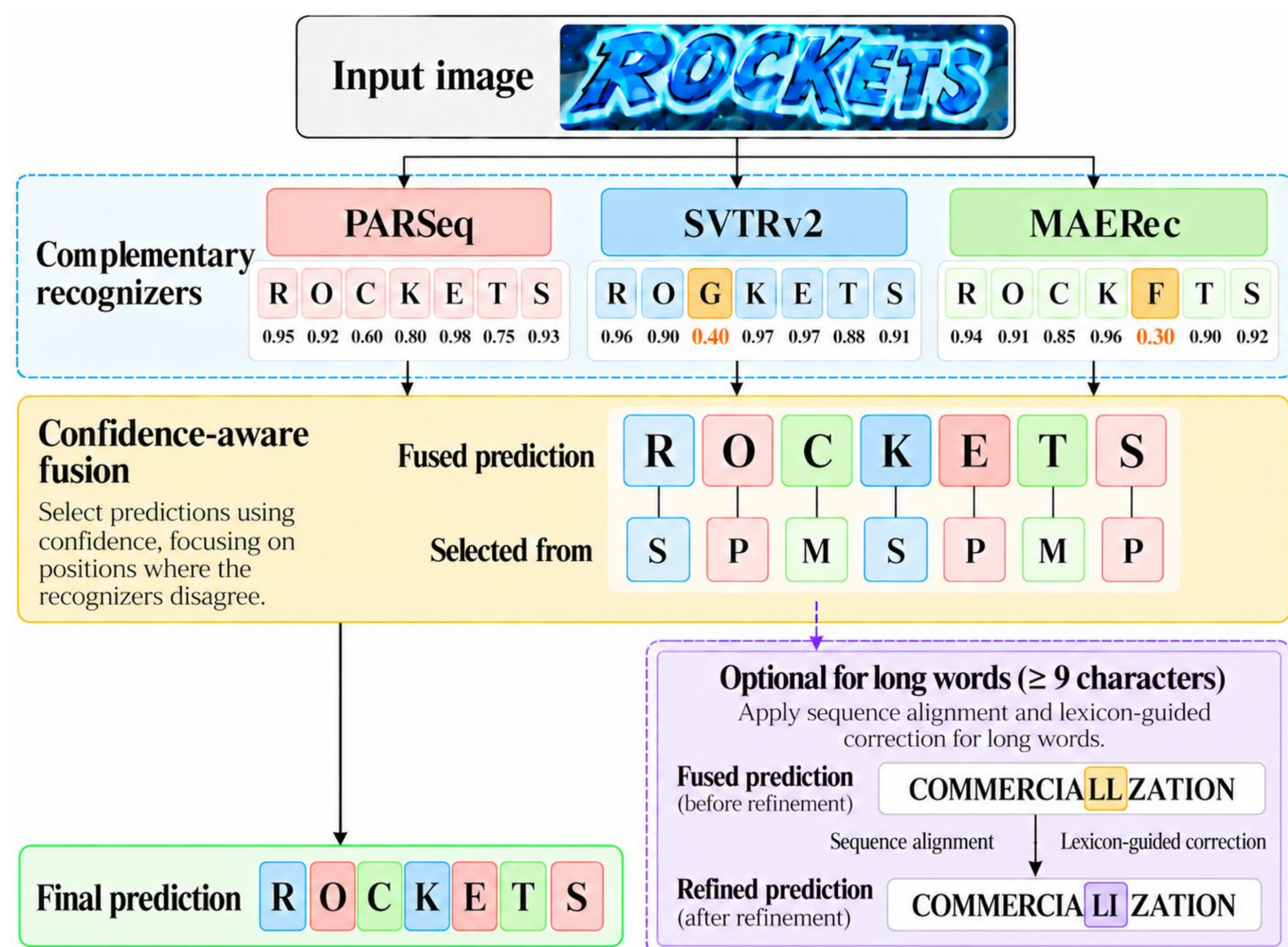


Figure 2. Overview of the proposed method.

Dataset

WordArt-V1.5: 12,000 artistic text images: 6,000 for training, 3,000 for Test A, and 3,000 for Test B.

Protocol in our work: Test A is used for validation and model selection, while Test B is reserved for final evaluation.



Figure 3. Examples from the WordArt-V1.5 dataset.

Results

Table 1. Ablation study on ensemble models.

Model	WRA (%)
MAERec	87.50
PARSeq	87.60
SVTRv2	88.13
Ensemble	89.90

Table 2. Ablation study on ensemble refinement for **long words (9+ characters)**.

Refinement	WRA (%)
None	75.00
Needleman-Wunsch only	75.00
Dictionary	76.63
Needleman-Wunsch + dictionary	77.72

Qualitative Results



PARSeq: cherry (0.94)
SVTRv2: chenny (0.89)
MAERec: champag (0.74)
ENSEMBLE: cherry (0.94)
LABEL: cherry



PARSeq: faherine (0.79)
SVTRv2: echevins (0.83)
MAERec: feberias (0.72)
ENSEMBLE: echevins (0.83)
LABEL: echevins

Error Analysis

Labeling error



PARSeq: vanilla (0.96)
SVTRv2: vanilla (0.96)
MAERec: vanilla (0.95)
ENSEMBLE: vanilla (0.96)
LABEL: promotion

Ambiguity



PARSeq: seller (0.95)
SVTRv2: seller (0.90)
MAERec: seller (0.82)
ENSEMBLE: seller (0.95)
LABEL: setter

Illegible



PARSeq: hangelou (0.70)
SVTRv2: hage (0.77)
MAERec: hangebou (0.73)
ENSEMBLE: hage (0.77)
LABEL: hangshu

Conclusions

- Proposed approach reached **89.90%**, outperforming the best fine-tuned model by **1.77 percentage points (pp)**.
- Long-word refinement brought a modest overall gain, with **+2.72 pp** gain on the targeted subset.
- Error analysis showed that **48.8%** of errors were linked to labeling issues, ambiguity, or illegibility.

Acknowledgments



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